Numerical Study of Vortex Shedding Behind a Circular Cylinder

Unsteady aerodynamic forces and formation of a Kármán vortex street

1. Aim

The objective of this experiment is to simulate the unsteady flow past a circular cylinder and observe the formation of alternating vortices in its wake. The oscillating lift response is used to obtain an approximate vortex-shedding frequency and Strouhal number.

2. Computational Setup

The flow is represented in a cylindrical coordinate system and solved using the D2Q9 BGK lattice Boltzmann model. The cylinder boundary is treated with a bounce-back scheme. Hydrodynamic forces are evaluated through momentum exchange, and the shedding behaviour is assessed from the time histories of lift and drag.

Simulation parameter	Value
Coordinate system	Cylindrical
Radial grid points	80
Azimuthal grid points	160
Reynolds number (main setup)	500
Total iterations	1200
Lattice model	D2Q9 BGK Lattice Boltzmann

3. Boundary Conditions

A low-Mach-number velocity is prescribed at the inlet, while an open boundary is used at the outlet. Wall treatment is applied along the upper and lower boundaries. The no-slip condition at the cylinder surface is imposed using bounce-back boundary treatment.

4. Quantities Monitored

The analysis considers the lift and drag coefficient histories, mean drag after the initial transient, lift amplitude, estimated shedding period and Strouhal number. Density deviation and vorticity contours are also examined to assess the wake pattern and its periodic development.

5. Results — Main Reported Case

The first results summary in the supplied report gives the following approximate values for the Reynolds-number-500 case.

Quantity	Approximate value
Maximum lift coefficient	+0.21
Minimum lift coefficient	–0.15
Lift peak-to-peak range	0.36
Initial maximum drag coefficient	20
Developed mean drag coefficient	1.5–1.6
Vortex-shedding period	7–8 (dimensionless time)
Estimated Strouhal number	0.13–0.14
Vorticity range	–9 to +7.5

6. Additional Results — Reynolds Number 510

A later section of the source report presents a second set of plots and values at Reynolds number 510. These results are kept separate from the main Reynolds-number-500 summary.

Quantity	Approximate value
Maximum positive lift coefficient	≈ +1.5
Minimum lift coefficient	≈ –1.6
Lift peak-to-peak range	≈ 3.1
Initial maximum drag coefficient	≈ 19
Developed mean drag coefficient	≈ 1.5–2.5
Vortex-shedding period	≈ 5–7
Estimated Strouhal number	≈ 0.14–0.20
Vorticity range	≈ –10 to +10

7. Discussion

The lift-coefficient history in the main case shows alternating positive and negative fluctuations, which are associated with vortices being released from opposite sides of the cylinder. The drag signal contains a pronounced initial peak and then approaches a comparatively steady level. The vorticity contour shows alternating regions of opposite sign downstream, forming the characteristic wake pattern known as a Kármán vortex street.

For the additional Reynolds-number-510 case, the lift oscillations become much larger in the plotted interval. The corresponding vorticity field also displays a sequence of alternating structures farther downstream. The force histories therefore indicate a strongly unsteady wake in this case.

The shedding period and Strouhal number are estimated from the spacing between successive lift peaks. The source report describes these values as approximate because the oscillations are still developing near the end of the simulated time interval. A more accurate estimate would require extracting peak locations from the numerical force data.

8. Conclusion

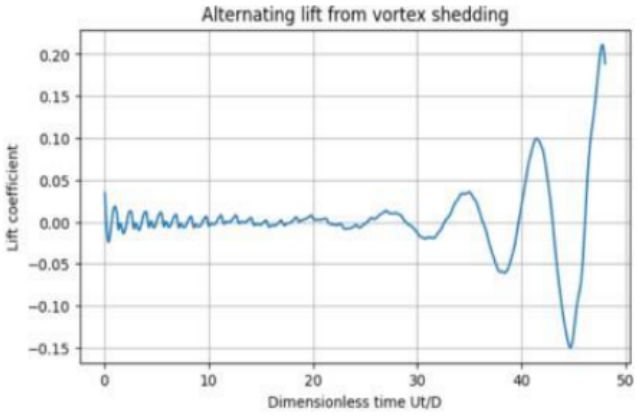
The numerical investigation demonstrates the formation of an unsteady wake behind a circular cylinder. Alternating lift fluctuations and the downstream vorticity pattern are consistent with periodic vortex shedding. The drag history exhibits a large initial transient followed by a lower developed level.

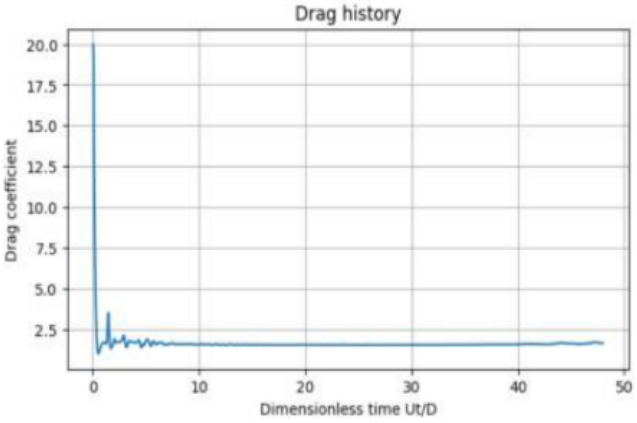
The supplied report contains two result sets: one identified with Reynolds number 500 and another with Reynolds number 510. Their lift amplitudes, drag ranges and estimated Strouhal numbers differ, so they have been presented independently rather than combined. The shedding-period and Strouhal estimates should be verified using the original force-history data before being treated as final quantitative values.

9. Original Simulation Figures

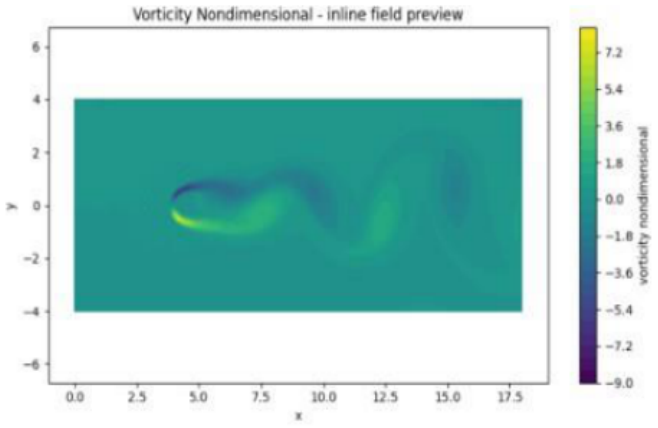
The original lift and drag histories, vorticity contours and supplementary result plots are reproduced in the following pages.

Original simulation output (source page 2)





Original simulation output (source page 3)

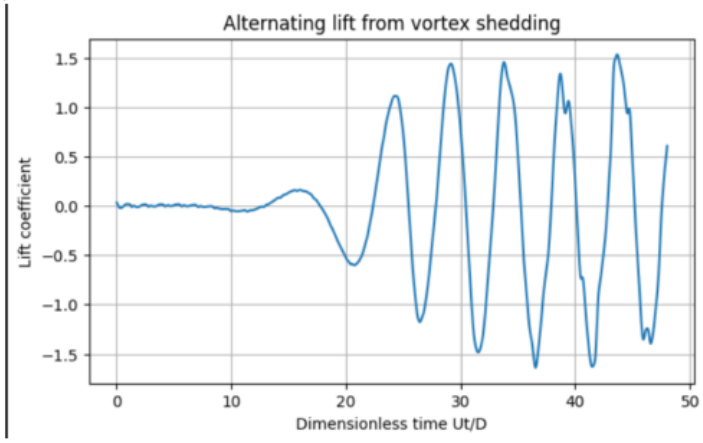


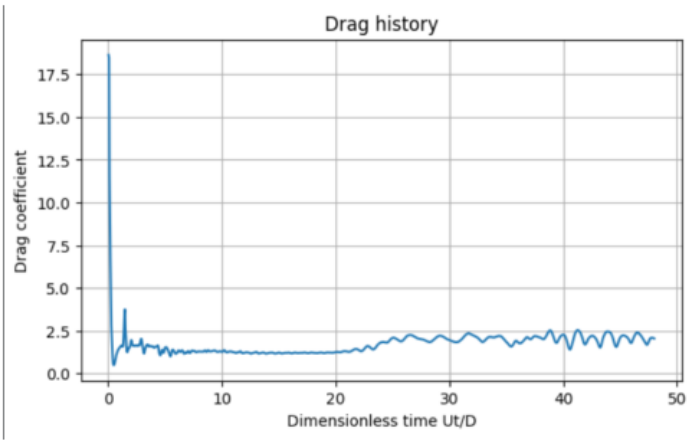
Results:

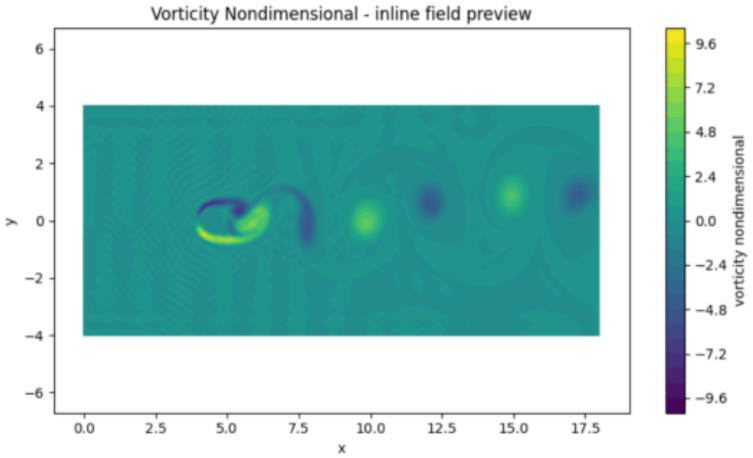
Parameter	Result / Approx. Value
Reynolds number	500
Grid	80 × 160
Total iterations	1200
Maximum positive lift coefficient,	≈ +0.21
Minimum lift coefficient,	≈ −0.15
Lift coefficient peak-to-peak	≈ 0.36
Initial maximum drag coefficient	≈ 20
Developed mean drag coefficient	≈ 1.5–1.6
Approx. vortex shedding period,	≈ 7–8
Estimated Strouhal number,	≈ 0.13–0.14
Vorticity range	≈ −9 to +7.5

Original simulation output (source page 4)

For RE = 510







Original simulation output (source page 5)

Parameter	Result / Approx. Value
Reynolds number	510
Grid	80 × 160
Total iterations	1200
Maximum positive lift coefficient	≈ +1.5
Minimum lift coefficient	≈ −1.6
Lift coefficient peak-to-peak	≈ 3.1
Initial maximum drag coefficient	≈ 19
Developed mean drag coefficient	≈ 1.5–2.5
Approx. vortex shedding period	≈ 5–7
Estimated Strouhal number	≈ 0.14–0.20
Vorticity range	≈ −10 to +10

Results and Discussion

The numerical simulation of vortex shedding behind a circular cylinder was performed at a Reynolds number of 510 using a computational grid of 80 × 160 and 1200 iterations. The lift coefficient history shows alternating positive and negative fluctuations, indicating the formation of an unsteady wake behind the cylinder.

From the plotted lift history, the maximum positive lift coefficient is approximately +1.5, while the minimum value is approximately −1.6. The peak-to-peak lift coefficient is therefore approximately 3.1. The drag coefficient initially reaches a high value of approximately 19 during the transient stage and subsequently fluctuates around an estimated range of 1.5–2.5.

The vorticity contour illustrates alternating regions of positive and negative vorticity downstream of the cylinder, which are characteristic of vortex shedding. The observed fluctuations in lift and drag indicate the unsteady aerodynamic forces associated with the wake.

The vortex shedding period and Strouhal number should be determined from successive peaks in the lift coefficient history. The values estimated visually from the available graphs are preliminary and require confirmation using the numerical output.

Conclusion

The flow around a circular cylinder at a Reynolds number of 510 was investigated numerically. The lift coefficient history showed alternating oscillations, while the drag coefficient exhibited a large initial transient followed by smaller fluctuations. The vorticity field displayed alternating positive and negative structures in the downstream wake, indicating vortex shedding.

The approximate maximum and minimum lift coefficients were +1.5 and −1.6, respectively, giving a peak-to-peak range of approximately 3.1. The drag coefficient fluctuated around an estimated range of 1.5–2.5 after the initial transient.

The results demonstrate the unsteady nature of the flow and the associated variation in aerodynamic forces. However, the shedding period and Strouhal number require quantitative analysis of the lift history, and the approximate graph-based values should be verified against the solver output before final submission.